Points 6 Published

Details Questions

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Question 0.55 pts

In a D/d cell, D and d are on:

- homologs
- non-homologous chromosomes
- sister chromatids
- different strands of DNA

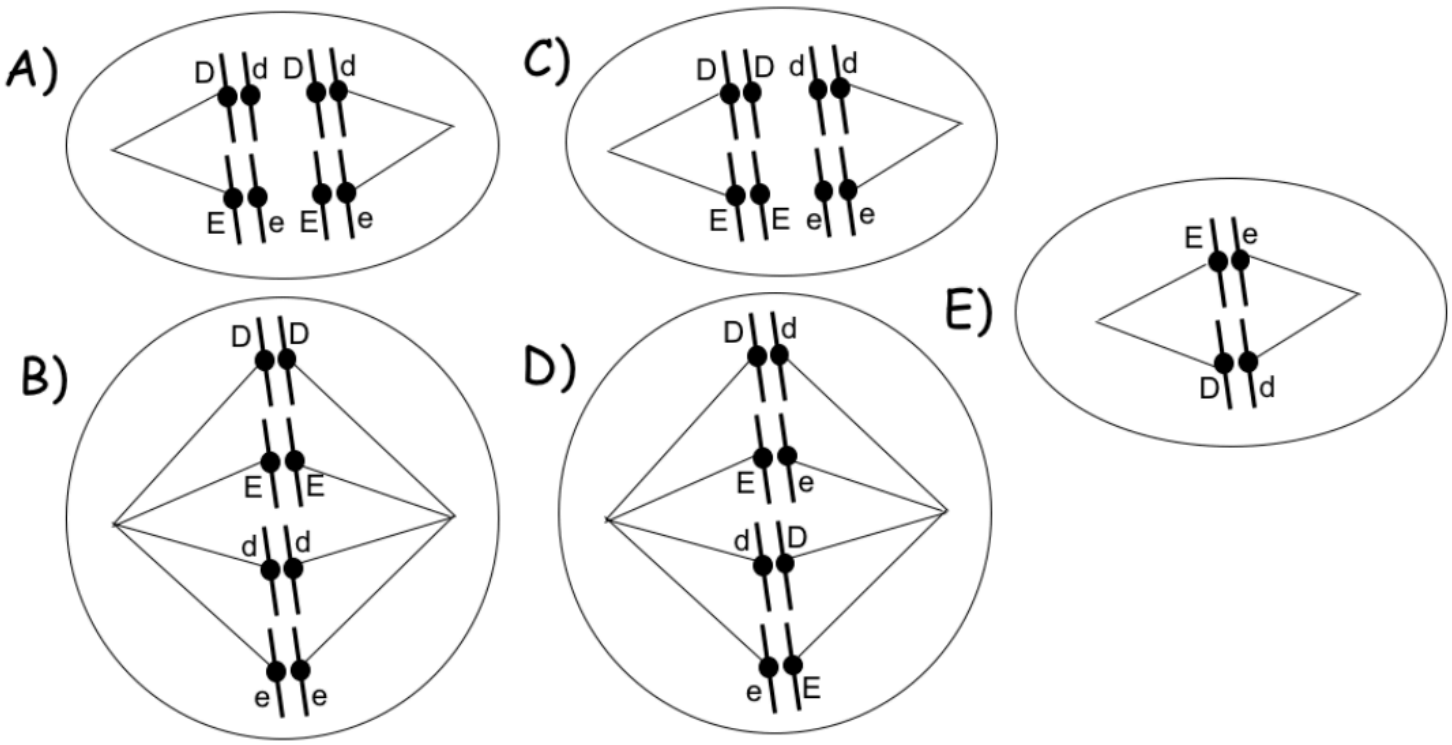
Question 0.59 pts

A D/d cell is:

1N
2N
3N
4N

Question 0.81 pts

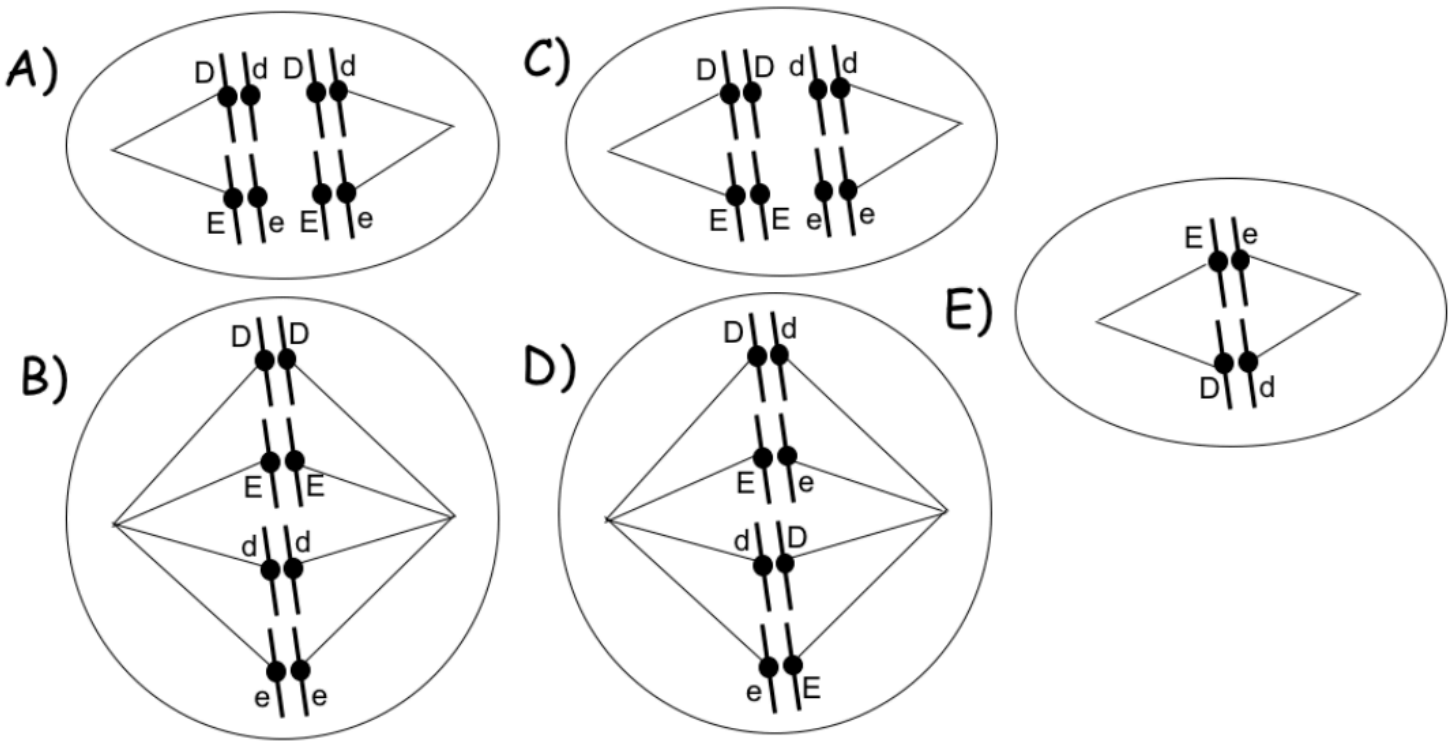
For a D/d; E/e cell, which diagram accurately represents mitosis?



A
B
C
D
E

Question 0.81 pts

For a D/d; E/e cell, which diagram accurately represents **meiosis I**?



A
B
C
D
E

Question 0.81 pts

For a D/d; E/e organism, which are the the correct genotypes for the 4 products of meiosis from a <u>single</u> germ cell, if all goes well and there are no cross-overs? It might be good to draw out the possible steps of meiosis.

D;e D;e d;E d;E
D;e d;E D;E d;e
D;d D;d E;e E;e
D d E e

Question 0.81 pts

Start with a single E/e; G/g diploid germ cell. It suffers nondisjunction of the E and e chromatids during **meiosis I**. That means the cell fails to correctly segregate those homologs at meiosis I. Both go to 1 spindle pole. The G an g chromatids disjoin fine. List the genotypes (at the E and G locus) of 4 gametes that could be produced by this meiosis. Assume there are no cross-overs.

E/e;g E/e;g G G
E/E;G e/e;G g g
E/e;G E/e;g g g
E/E;G e/e;g g G

Question 0.81 pts

Start with a single E/e; G/g diploid germ cell. It suffers nondisjunction of the E sister chromatids during **meiosis II**. That means it fails to correctly segregate those sisters at meiosis II. Both go to 1 spindle pole. The G an g chromatids disjoin fine. List the genotypes (at the E and G locus) of 4 gametes that could be produced by this meiosis. Assume there are no cross-overs.

E/e;g E/e;g G G
E/E;G G e;g e;g
E/E;G e/e;G g g
E/E;G G e/e;g g

Question 0.81 pts

Alleles for the human gene *DYNLL2* would be inherited independently of alleles in the human gene *HSPA9.* Hint: Search for the chromosome location of these genes in the Human GRCh38/hg38 assembly in the UCSC Genome Browser (**genome.ucsc.edu (http://genome.ucsc.edu)**).

True
False

+ New Question
+ New Question Group
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